

Final pstat 126

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Part 1

1.1 Selecting random sample

```
library(car)
```

```
## Loading required package: carData
```

```
diamonds <- read.csv("~/Downloads/Diamonds Prices2022.csv")
```

```
#1.1 Select random sample
```

```
summary(diamonds)
```

```
##           X           carat           cut           color
## Min.      :    1   Min.    :0.2000   Length:53943   Length:53943
## 1st Qu.:13486   1st Qu.:0.4000   Class :character   Class :character
## Median :26972   Median :0.7000   Mode  :character   Mode  :character
## Mean    :26972   Mean    :0.7979
## 3rd Qu.:40458   3rd Qu.:1.0400
## Max.    :53943   Max.    :5.0100
## clarity     depth           table           price
## Length:53943   Min.    :43.00   Min.    :43.00   Min.    : 326
## Class :character 1st Qu.:61.00   1st Qu.:56.00   1st Qu.: 950
## Mode  :character Median :61.80   Median :57.00   Median : 2401
##                  Mean  :61.75   Mean  :57.46   Mean  : 3933
##                  3rd Qu.:62.50   3rd Qu.:59.00   3rd Qu.: 5324
##                  Max.   :79.00   Max.   :95.00   Max.   :18823
##           x           y           z
## Min.      : 0.000   Min.    : 0.000   Min.    : 0.000
## 1st Qu.: 4.710   1st Qu.: 4.720   1st Qu.: 2.910
## Median : 5.700   Median : 5.710   Median : 3.530
## Mean    : 5.731   Mean    : 5.735   Mean    : 3.539
## 3rd Qu.: 6.540   3rd Qu.: 6.540   3rd Qu.: 4.040
## Max.    :10.740   Max.    :58.900   Max.    :31.800
```

```
set.seed(123)
```

```
Rsample <- sample(1:nrow(diamonds), 1000,
                  replace = FALSE)
```

```
diamondsample<-diamonds[Rsample, ]
```

1.2 Describe the variables

```
summary(diamondsample)
```

```
##           X           carat           cut           color
## Min.      : 31   Min.    :0.2300   Length:1000   Length:1000
## 1st Qu.:13223   1st Qu.:0.4000   Class :character   Class :character
## Median :26435   Median :0.7100   Mode  :character   Mode  :character
## Mean    :26658   Mean    :0.7962
## 3rd Qu.:40271   3rd Qu.:1.0400
## Max.    :53912   Max.    :2.3200
## clarity    depth           table           price
## Length:1000   Min.    :57.50   Min.    :52.00   Min.    : 368.0
## Class :character   1st Qu.:61.00   1st Qu.:56.00   1st Qu.: 956.8
## Mode  :character   Median :61.90   Median :57.00   Median : 2522.0
##                      Mean    :61.79   Mean    :57.41   Mean    : 3935.9
##                      3rd Qu.:62.50   3rd Qu.:59.00   3rd Qu.: 5301.8
##                      Max.    :69.60   Max.    :66.00   Max.    :18706.0
##           x           y           z
## Min.    :3.920   Min.    :3.960   Min.    :0.000
## 1st Qu.:4.707   1st Qu.:4.718   1st Qu.:2.900
## Median :5.720   Median :5.720   Median :3.540
## Mean    :5.733   Mean    :5.736   Mean    :3.539
## 3rd Qu.:6.530   3rd Qu.:6.513   3rd Qu.:4.030
## Max.    :8.570   Max.    :8.520   Max.    :5.280
```

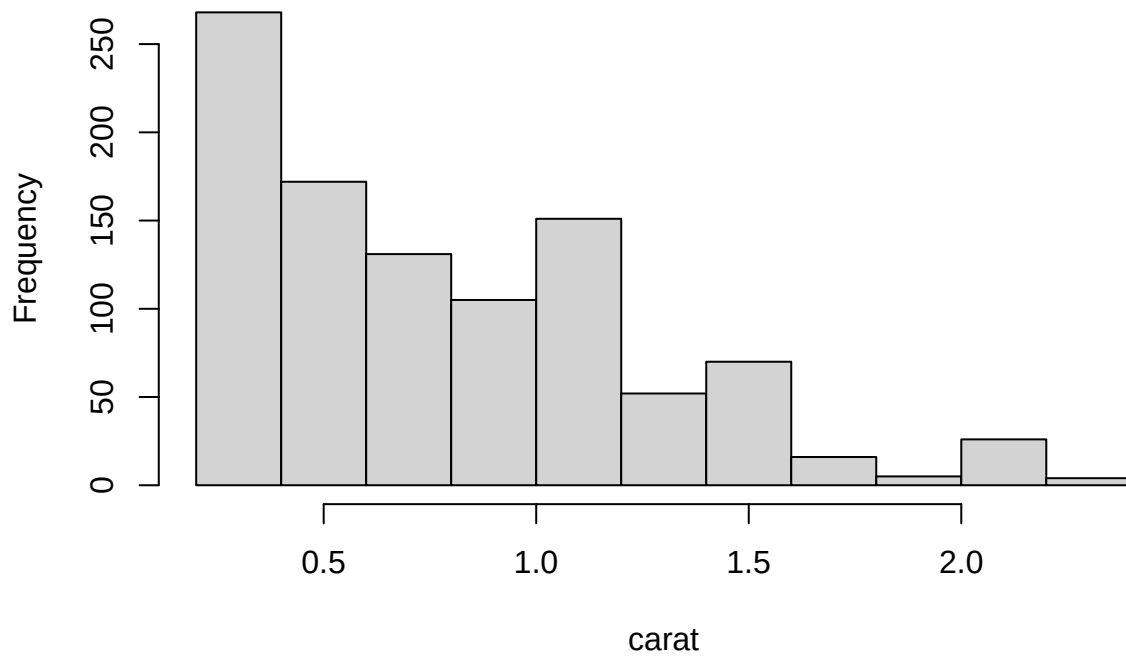
```
str(diamondsample)
```

```
## 'data.frame': 1000 obs. of 11 variables:
## $ X : int 51663 2986 29925 29710 37529 2757 45404 46435 38938 9642 ...
## $ carat : num 0.73 0.7 0.31 0.31 0.31 0.83 0.51 0.7 0.4 1.1 ...
## $ cut : chr "Ideal" "Ideal" "Ideal" "Ideal" ...
## $ color : chr "I" "G" "D" "H" ...
## $ clarity: chr "VS1" "VS1" "VS1" "VVS1" ...
## $ depth : num 60.7 60.8 61.6 62.2 60.9 63.7 62.5 64.2 61.6 61.2 ...
## $ table : num 56 56 55 56 55 59 58 58 56 61 ...
## $ price : int 2397 3300 713 707 987 3250 1668 1771 1053 4640 ...
## $ x : num 5.85 5.73 4.3 4.34 4.39 5.95 5.12 5.59 4.73 6.61 ...
## $ y : num 5.81 5.8 4.33 4.37 4.41 5.89 5.18 5.62 4.78 6.66 ...
## $ z : num 3.54 3.51 2.66 2.71 2.68 3.77 3.22 3.6 2.93 4.01 ...
```

Histograms RV

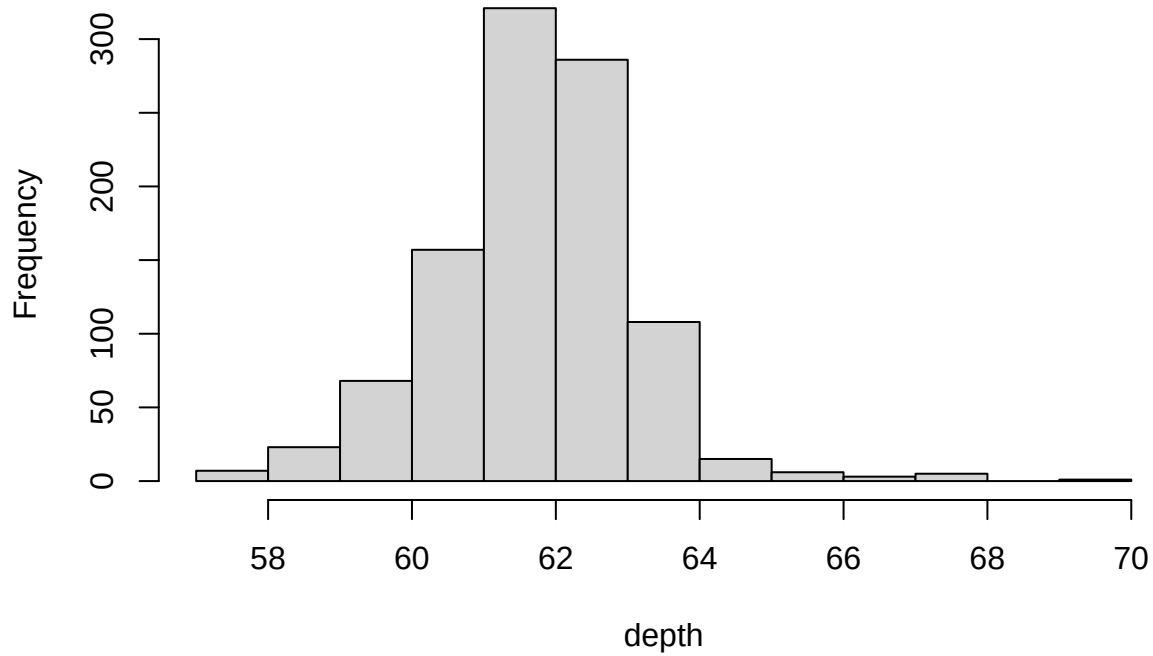
```
hist(diamondsample$carat,main="Histogram of carat",xlab="carat")
```

Histogram of carat



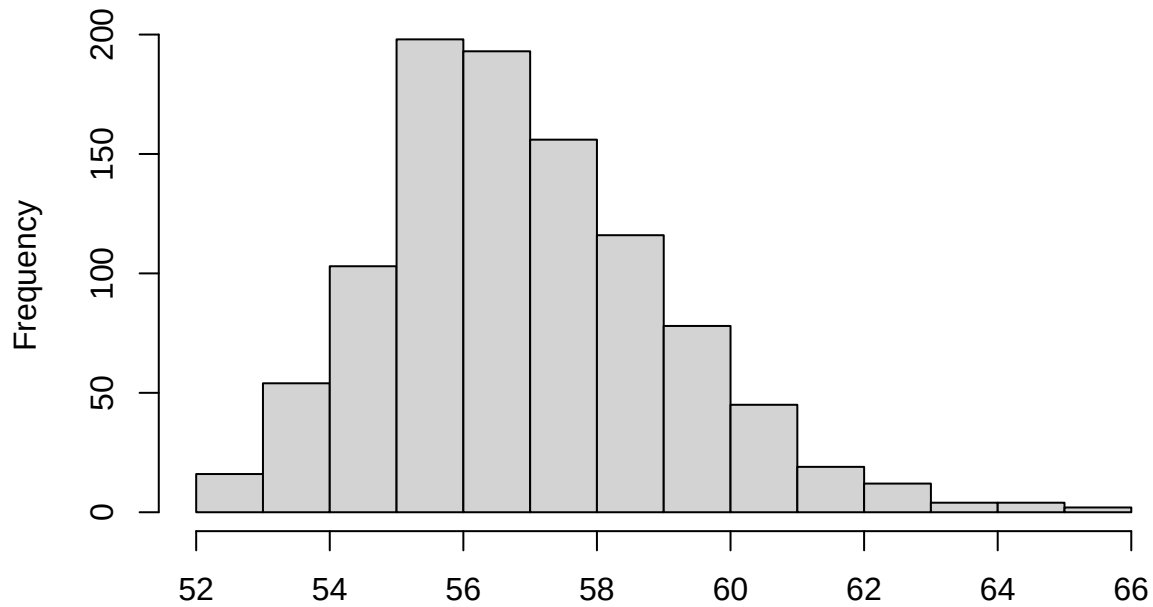
```
hist(diamondsample$depth,main="Histogram of Depth",xlab="depth")
```

Histogram of Depth



```
hist(diamondsample$table,main="Histogram of table",xlab="Table")
```

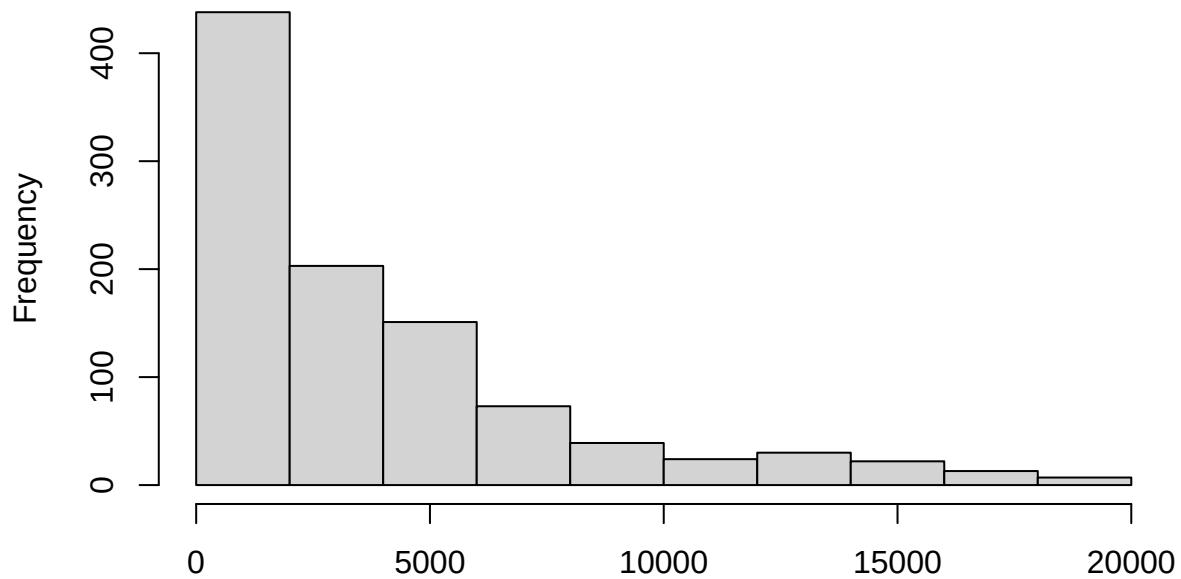
Histogram of table



Table

```
hist(diamondsample$price,main="Histogram of price",xlab="price")
```

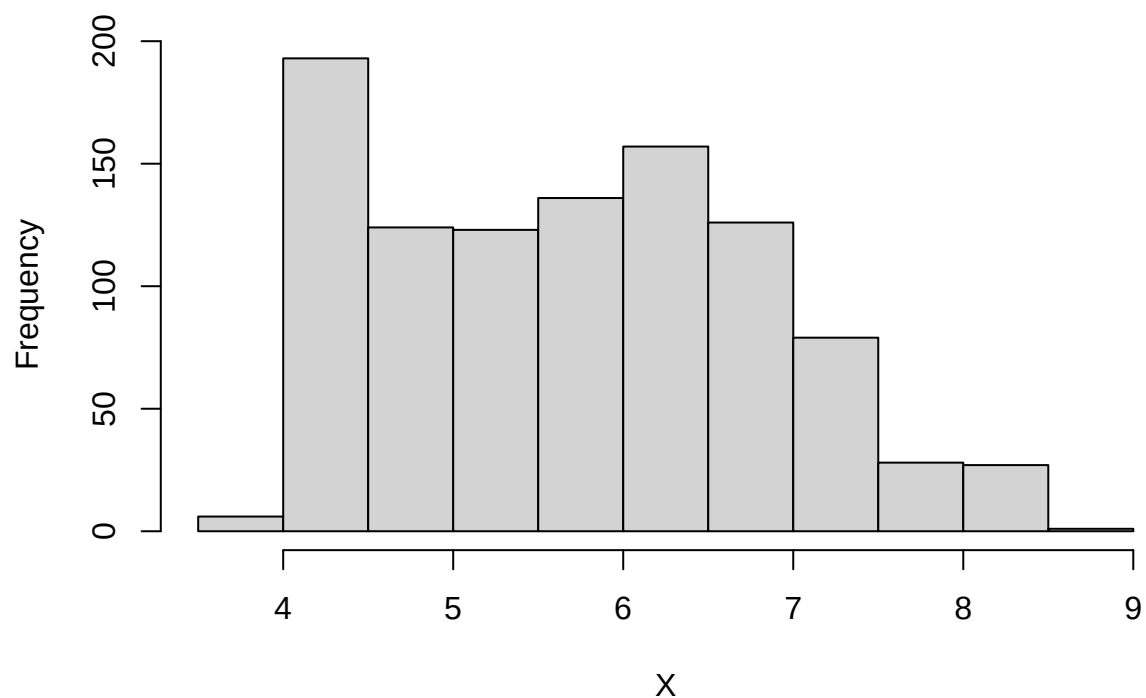
Histogram of price



price

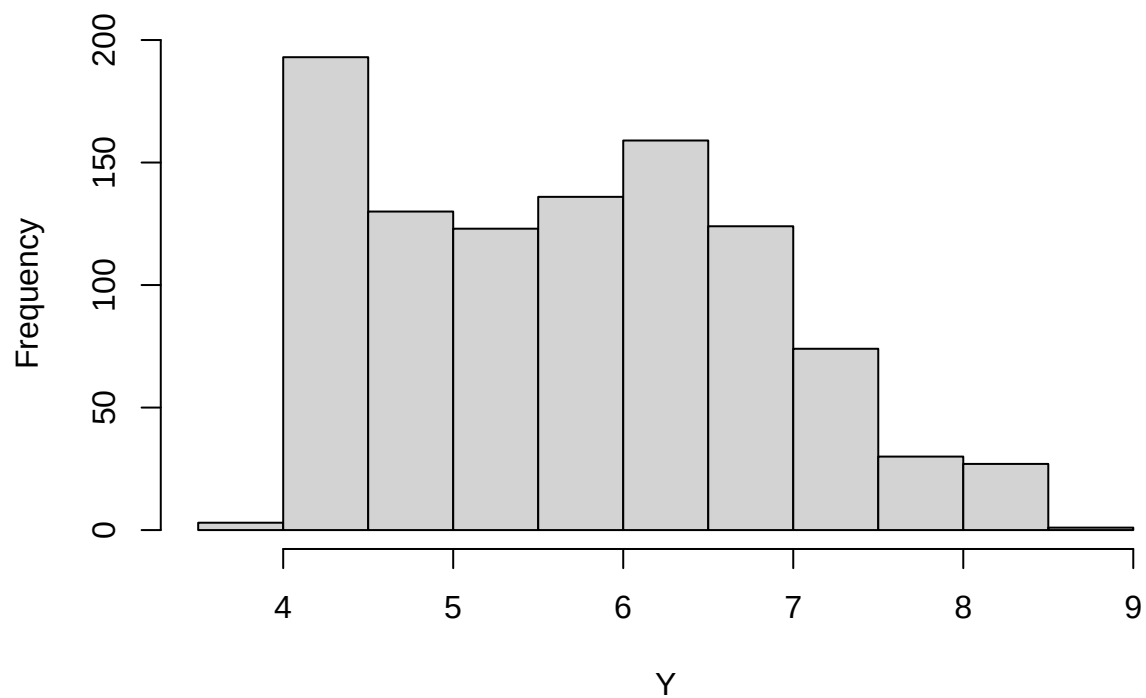
```
hist(diamondsample$x,main="Histogram of X",xlab="X")
```

Histogram of X

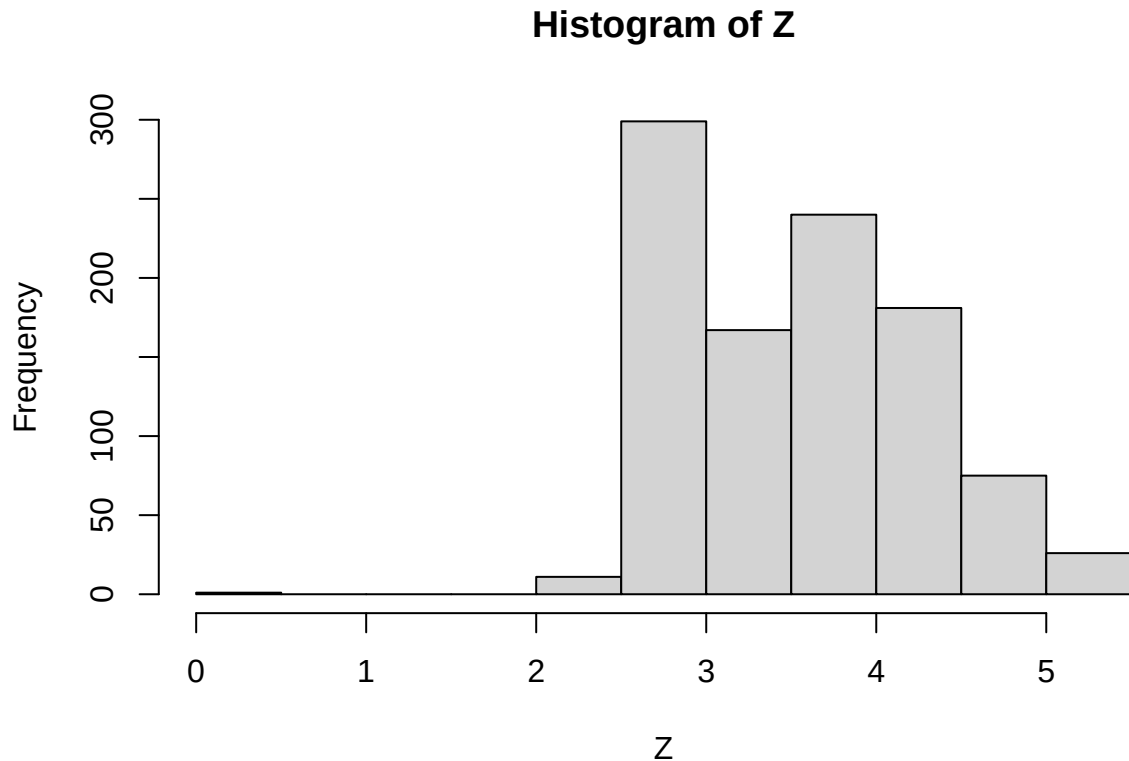


```
hist(diamondsample$y,main="Histogram of Y",xlab="Y")
```

Histogram of Y



```
hist(diamondsample$z,main="Histogram of Z",xlab="Z")
```



The histogram for carat shows a right skewed distribution. Most diamonds have a smaller carat value with higher ones having less frequency.

We can see that the data for depth is relatively symmetric and centered in the mid 60s. There is a slight skew to the right which tells us there is not much variation.

The histogram of table is skewed to the right with high frequency in the 55–58 range.

The price distribution is right skewed as the more expensive the less frequency there is. This tells us that there are less expensive diamonds out there.

The histogram of X has similar frequency from 4 to 7 but then drops significantly after. This tells us that larger diamonds are less frequent.

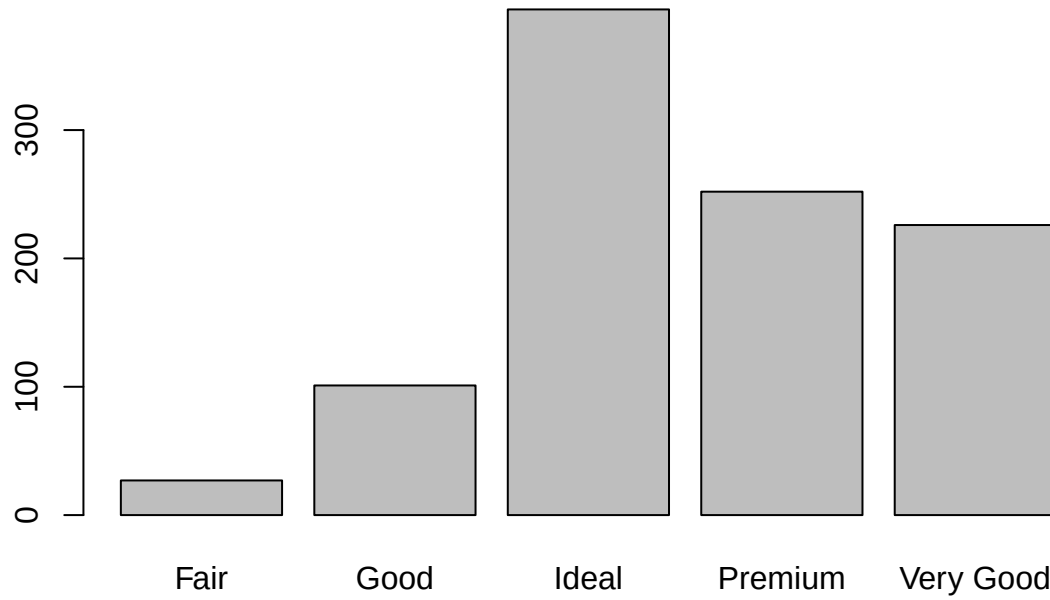
The histogram of Y is very similar to that of X with only slight differences in the values. This tells us that like X, there are less larger diamonds.

The histogram of Z has strong frequency at 2.75 and at 4 but is relatively low despite those values. This tells us that most diamonds do not have extreme values.

BarPlot For Categorical Variables

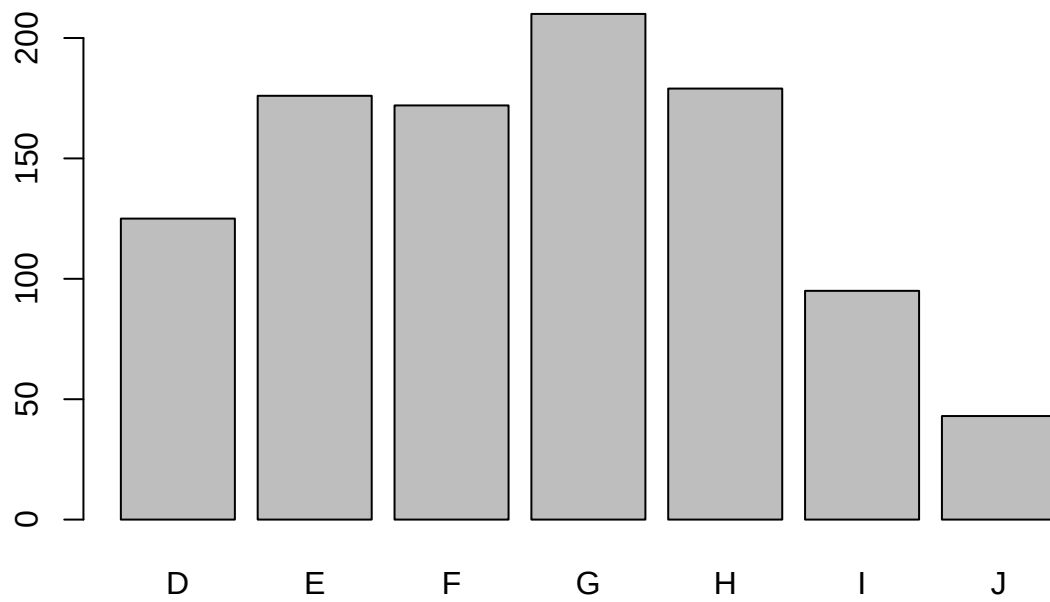
```
barplot(table(diamondsample$cut),main="Cut distribution")
```

Cut distribution

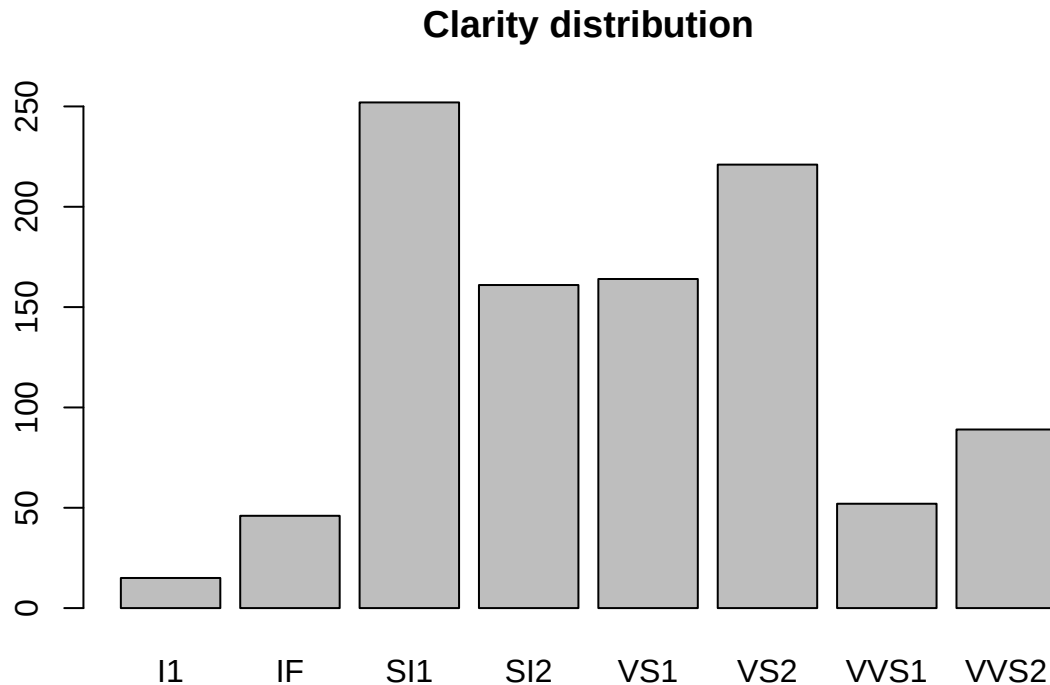


```
barplot(table(diamondsample$color),main="Color distribution")
```

Color distribution



```
barplot(table(diamondsample$clarity),main="Clarity distribution")
```



The cut with the highest frequency is ideal, with premium and good also having high frequency. We can also see that fair and good are not as frequent.

The color distribution is centered around G with I and J having the lowest.

The clarity has highest frequency with SI1 and VS2 with the lowest and highest values having lowest frequency.

1.3 checking for correlation

```
VarCheck <-diamondsample[, c("carat","depth","table","price","x","y","z")]
cor(VarCheck)
```

	carat	depth	table	price	x	y	z
carat	1.0000000	0.052779405	0.2114770	0.923340431	0.980725752	0.980857857	0.9644749
depth	0.0527794	1.000000000	-0.2566595	0.009338803	0.004846103	0.001814092	0.1377042
table	0.2114770	-0.256659487	1.0000000	0.147609194	0.233127900	0.230644626	0.1932476
price	0.9233404	0.009338803	0.1476092	1.000000000	0.881271345	0.884552920	0.8669840
x	0.9807258	0.004846103	0.2331279	0.881271345	1.000000000	0.998950088	0.9747044
y	0.9808579	0.001814092	0.2306446	0.884552920	0.998950088	1.000000000	0.9746257
z	0.9644749	0.137704234	0.1932476	0.866983958	0.974704370	0.974625715	1.0000000

When running the correlation matrix, I saw a strong relationship with price and size related variables such as carat, x, y, z. We can see that Price and Carat have a correlation of .93 which tell us larger carat sized diamonds are more expensive. We can also see for example that depth and table show weak correlation with price meaning that they have less influence on price as opposed to size variables.

1.4 Multiple Linear Regression

```
MLRmodel <- lm(price ~ carat + color + cut + clarity + depth + table + x + y
               + z, data = diamonds.sample)
summary(MLRmodel)
```

```
##
## Call:
## lm(formula = price ~ carat + color + cut + clarity + depth +
##     table + x + y + z, data = diamonds.sample)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4375.3  -563.2  -177.1   392.9  6463.4
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   9101.25    3191.77   2.851 0.004443 **
## carat        14152.49    405.88  34.869 < 2e-16 ***
## colorE       -250.14    126.67  -1.975 0.048575 *
## colorF       -240.34    127.57  -1.884 0.059862 .
## colorG       -543.08    124.05  -4.378 1.33e-05 ***
## colorH      -1057.16    128.37  -8.235 5.72e-16 ***
## colorI      -1660.84    150.62 -11.027 < 2e-16 ***
## colorJ      -2467.68    195.13 -12.646 < 2e-16 ***
## cutGood        308.22    268.35   1.149 0.251000
## cutIdeal       565.84    272.97   2.073 0.038443 *
## cutPremium     354.90    261.74   1.356 0.175441
## cutVery Good   502.34    264.37   1.900 0.057711 .
## clarityIF     3889.34    359.43  10.821 < 2e-16 ***
## claritySI1    2756.88    321.59   8.573 < 2e-16 ***
## claritySI2    1804.18    326.61   5.524 4.25e-08 ***
## clarityVS1    3789.18    327.79  11.560 < 2e-16 ***
## clarityVS2    3529.03    325.62  10.838 < 2e-16 ***
## clarityVVS1   3939.78    355.25  11.090 < 2e-16 ***
## clarityVVS2   4012.24    339.65  11.813 < 2e-16 ***
## depth        -79.38     36.78  -2.158 0.031153 *
## table         -38.56     20.96  -1.840 0.066077 .
## x            -2845.46    809.25  -3.516 0.000458 ***
## y             1003.05    810.83   1.237 0.216363
## z            -476.83    288.48  -1.653 0.098675 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1070 on 976 degrees of freedom
## Multiple R-squared:  0.9294, Adjusted R-squared:  0.9278
## F-statistic: 559 on 23 and 976 DF, p-value: < 2.2e-16
```

1.5

I thought that it was interesting to work with data specifically about diamonds. I would say for the most part that the data appeared to line up with I would have pictured when it came to quality the frequency revolved around the mean and I expected that.

Part 2

2.1 Simple linear regression

```
SLRmodel <- lm(price ~ x, data =diamondsample)
summary(SLRmodel)

##
## Call:
## lm(formula = price ~ x, data = diamondsample)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4777.5 -1295.5  -215.3   994.9  9170.0
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -14312.66     315.45  -45.37  <2e-16 ***
## x             3183.29       54.04   58.91  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1882 on 998 degrees of freedom
## Multiple R-squared:  0.7766, Adjusted R-squared:  0.7764
## F-statistic: 3470 on 1 and 998 DF, p-value: < 2.2e-16
```

2.2

confidence interval

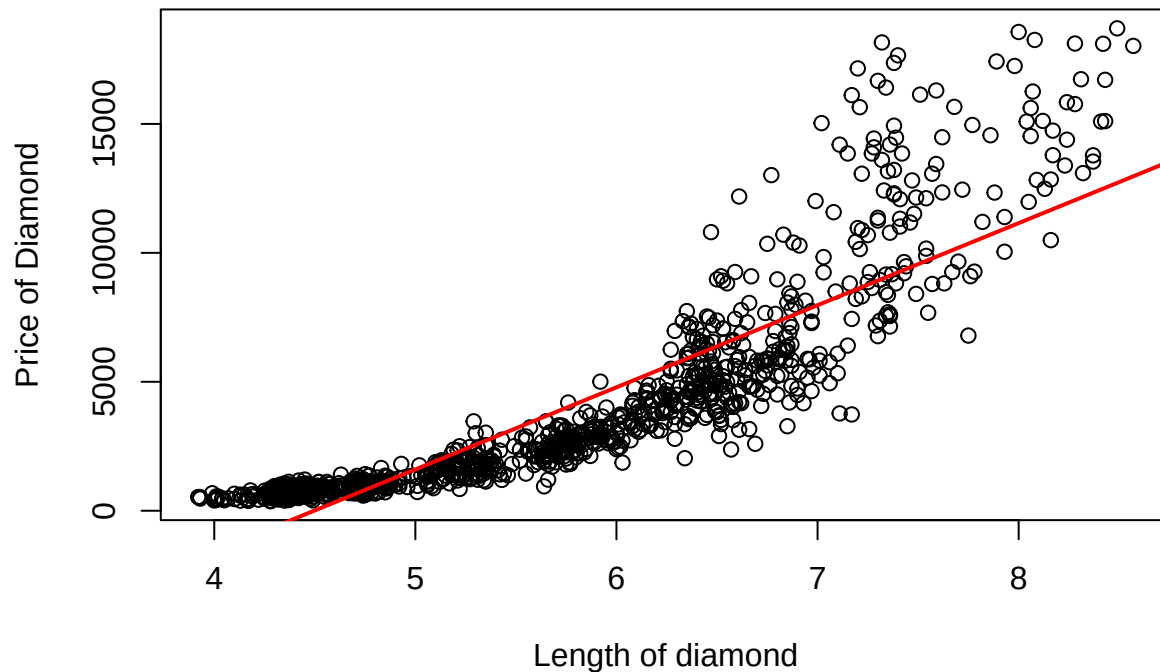
```
confint(SLRmodel)

##              2.5 %    97.5 %
## (Intercept) -14931.673 -13693.65
## x             3077.245   3289.33
```

Plotting the regression

```
plot(diamondsample$x, diamondsample$price,
      main = "Price vs diamond length", xlab = "Length of diamond", ylab = "Price of Diamond")
abline(SLRmodel, col = "red", lwd=2)
```

Price vs diamond length



Prediction interval

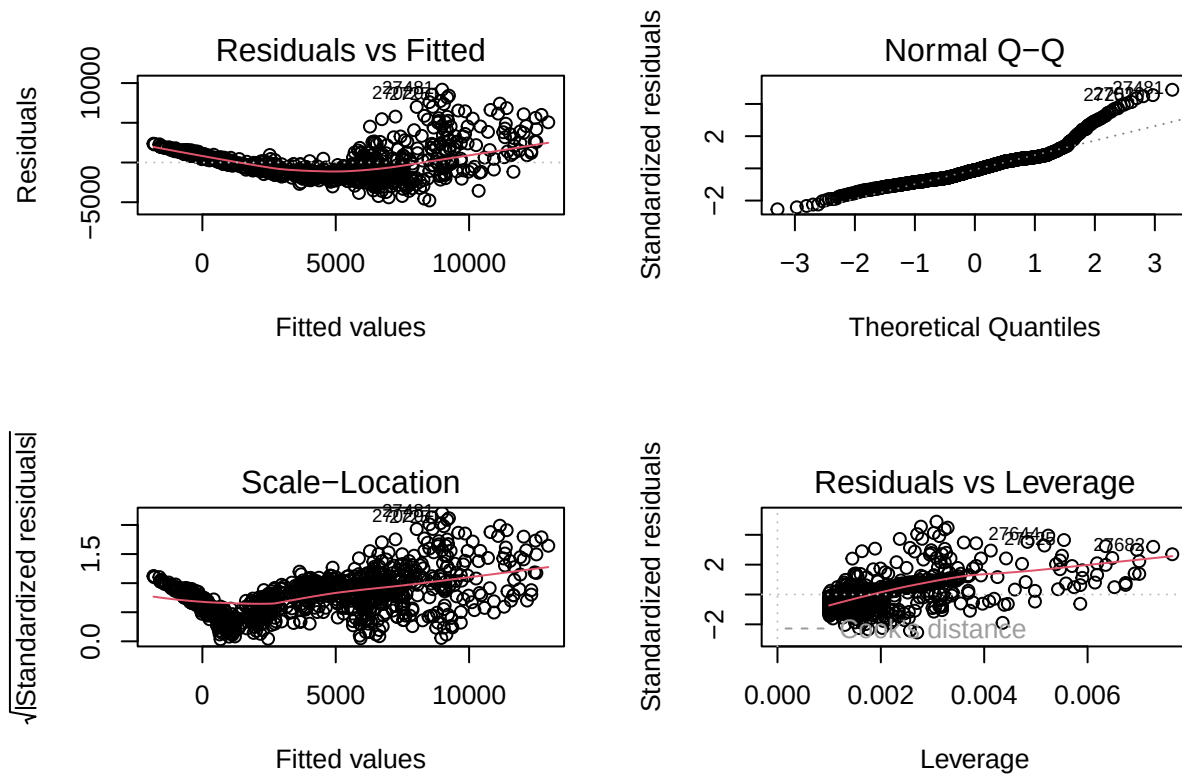
```
PredictData <-data.frame(x =5)
predict(SLRmodel,PredictData,interval ="prediction")
```

```
##          fit          lwr          upr
## 1 1603.776 -2091.671 5299.223
```

For my simple linear regression I decided to use diamond length as my predictor and price for my response variable. We can see a positive relationship as the larger the diamond, the higher the price generally. The model has an adj R^2 and R^2 value of .79 telling us about 79% of variation in diamond price can be explained by the length. We can see that every millimeter added increases the price by approximately 3131 units on average. For the prediction interval I chose to predict when length equals 5 the value is approximately 1589.93. The summary shows a p value of $2.2e^{-16}$, if we choose $p = .05$ we can see the value we got is very much smaller than .05 so we reject null hypothesis and say the length of diamond is a sufficient predictor of price. Our C.I tells us that we are .95% confident that adding 1 mm in length will add around 3031 to 3231 units of value.

2.3

```
par(mfrow=c(2,2))
plot(SLRmodel)
```



The plots helped us check the assumptions of the simple linear regression model. Residuals vs fitted showed us that the residuals are fairly randomly scattered. This tells us linearity is reasonable. The normal Q-Q plot tell us residuals are normally distributed with small deviations. Scale location plot tells us variance in residuals is constant. The residuals vs leverage plot does not show any influential observations. Overall regression assumptions appear satisfied.

2.4

```
Transformation <- lm(log(price) ~ x, data = diamondsamle)
summary(Transformation)

##
## Call:
## lm(formula = log(price) ~ x, data = diamondsamle)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.85867 -0.16754  0.01583  0.16725  0.84151
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.706484   0.044272   61.13  <2e-16 ***
## x             0.887122   0.007584  116.97  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2641 on 998 degrees of freedom
## Multiple R-squared:  0.932, Adjusted R-squared:  0.9319
## F-statistic: 1.368e+04 on 1 and 998 DF, p-value: < 2.2e-16
```

I transformed the response variable using $\log(\text{price})$. The transformed model showed a positive relationship with length and price. The log transformation helps improve model by making relationship more linear and reduces variability in residual.

2.5 Step 5: Additional predictors

I added predictors and see how it affected the adjusted R^2 . The original R^2 value was .7897. When I added Carat it increased the adjusted R^2 to .8677. This increase tells us that carat size also influences the price of diamonds. When I included depth the adjusted R^2 goes down slightly and does not explain the model's variation more. This tells us that diamond length is more important to the price than depth.

2.6

I was more comfortable working with simple linear regression as opposed to multiple. One thing that interested me was part 5 with adding different variables and seeing how the adjusted R^2 changed.

Part 3

3.1

```
MLRmodel <- lm(price ~ carat + cut + color + clarity + depth + table + x + y + z,
               data = diamonds)
stepMLR <- step(MLRmodel, direction = "both")
```

```
## Start: AIC=13973.61
## price ~ carat + cut + color + clarity + depth + table + x + y +
## z
##
##           Df Sum of Sq      RSS   AIC
## - y         1   1750432 1118137700 13973
## - cut        4    8768871 1125156140 13973
## <none>              1116387269 13974
## - z         1    3125012 1119512280 13974
## - table      1    3872442 1120259711 13975
## - depth      1    5327972 1121715240 13976
## - x         1   14141947 1130529216 13984
## - color      6   338474966 1454862234 14226
## - clarity    7   545022944 1661410213 14357
## - carat      1 1390732284 2507119553 14781
##
## Step: AIC=13973.18
## price ~ carat + cut + color + clarity + depth + table + x + z
##
##           Df Sum of Sq      RSS   AIC
## <none>              1118137700 13973
## - z         1    2671642 1120809343 13974
## + y         1   1750432 1116387269 13974
## - table      1    3765350 1121903050 13974
## - depth      1    6377623 1124515324 13977
## - cut        4   13245091 1131382791 13977
## - x         1   71564869 1189702570 14033
## - color      6   338842935 1456980635 14226
## - clarity    7   558485157 1676622857 14364
## - carat      1 1418316965 2536454666 14790
```

```
summary(stepMLR)
```

```
##
## Call:
## lm(formula = price ~ carat + cut + color + clarity + depth +
##     table + x + z, data = diamondsample)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4404.0  -554.1  -169.6   406.7  6367.1
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   9502.46    3176.11   2.992  0.00284 **
## carat        14207.17     403.57  35.204 < 2e-16 ***
## cutGood        370.51     263.65   1.405  0.16024
## cutIdeal       617.96     269.77   2.291  0.02220 *
## cutPremium     362.86     261.73   1.386  0.16595
## cutVery Good   577.75     257.32   2.245  0.02497 *
## colorE       -252.30     126.69  -1.991  0.04671 *
## colorF       -244.32     127.56  -1.915  0.05574 .
## colorG       -541.06     124.07  -4.361 1.43e-05 ***
## colorH      -1059.82     128.39  -8.255 4.90e-16 ***
## colorI      -1667.56     150.56 -11.075 < 2e-16 ***
## colorJ      -2464.75     195.17 -12.629 < 2e-16 ***
## clarityIF     3917.73     358.79  10.919 < 2e-16 ***
## claritySI1    2771.37     321.47   8.621 < 2e-16 ***
## claritySI2    1815.81     326.57   5.560 3.48e-08 ***
## clarityVS1    3814.83     327.22  11.658 < 2e-16 ***
## clarityVS2    3547.76     325.36  10.904 < 2e-16 ***
## clarityVVS1   3967.43     354.64  11.187 < 2e-16 ***
## clarityVVS2   4033.71     339.30  11.888 < 2e-16 ***
## depth        -85.94       36.41  -2.361  0.01844 *
## table        -38.01       20.96  -1.814  0.07001 .
## x           -1888.96     238.88  -7.908 7.06e-15 ***
## z            -438.31     286.88  -1.528  0.12687
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1070 on 977 degrees of freedom
## Multiple R-squared:  0.9293, Adjusted R-squared:  0.9277
## F-statistic: 584 on 22 and 977 DF,  p-value: < 2.2e-16
```

I ran the step wise regression with AIC. Essentially what this does is remove variables to improve the model and minimize AIC. The variable removed was y. By removing y from the model we were able to raise the adjusted R^2 to .93, improving the model.

3.2

```
vif(stepMLR)
```

```
##              GVIF Df GVIF^(1/(2*Df))
## carat      29.783176  1         5.457396
## cut        2.612825  4         1.127558
```

```
## color      1.281313  6      1.020872
## clarity    1.765683  7      1.041446
## depth      2.189661  1      1.479750
## table      1.820644  1      1.349312
## x          60.463794  1      7.775847
## z          34.463041  1      5.870523
```

I ran the vif function and when analyzing the results we can see that z and x, and carat have very high VIF values. This indicates strong multicollinearity between these predictors. Cut, color, and clarity all have VIF values close to 1, indicating little to no multicollinearity, meaning they are not strongly related to other predictors.

3.3

```
newdiamond <- data.frame( carat = 1 , cut = "Ideal", color ="G",
                           clarity = "VS2", depth = 61 ,table = 57, x = 6 , z=4)
predict(stepMLR,newdiamond,interval = "confidence")
```

```
##          fit          lwr          upr
## 1 6838.157 6538.986 7137.329
```

```
predict(stepMLR,newdiamond,interval = "prediction")
```

```
##          fit          lwr          upr
## 1 6838.157 4717.588 8958.726
```

I created a prediction model with the following values assigned to each variable. I had carat =1, cut = ideal , color = G,clarity = VS2, depth = 61, x= 6,and z =4. What I got was a predicted price of approximately 3650.80. The 95% C.I for mean predicted price was 2558.37 and 4743.24. This gives the average expected price of a diamond with the values I gave. The .95% P.I gave us a range from 1328.39 and 5973.21, the predicted price range for a future diamond with these characteristics. The P.I is wider because it takes into account uncertainty and variation in diamond prices.

3.4 Final summary

To begin this project I analyzed a dataset of diamond characteristics and the price. I selected a random sample of 1000 observations and viewed them in many different ways such as using summary statistics, histograms, and barplots. I found correlation in many of the size variables and the price of the diamonds. I then performed a simple linear regression where I used diamond length (x) as a predictor of price. I was able to confirm a strong relationship and the model explained that 79% of the variation in price. A hypothesis test then confirmed that length is a sufficient predictor of diamond price. I then created a multiple linear Regression model and tested assumptions then applied a log transformation. With stepwise regression with AIC, the model removed the y variable and produced an adjusted R^2 of .93. This told me that the new model explained a large portion of the variability in the price of diamonds. VIF then showed me strong multicollinearity in size variables x, z and carat which is expected as they describe dimensions of the diamond. Finally I calculated Confidence intervals and Prediction Intervals for future diamonds using my final model. The prediction interval showed to be wider as it took into account uncertainty. Overall the analysis showed that diamond size plays the most important role in determining the price of a diamond.